

Chapter 2

PCR & ELECTROPHORESIS

Contents

I. PCR – Polymerase Chain Reaction

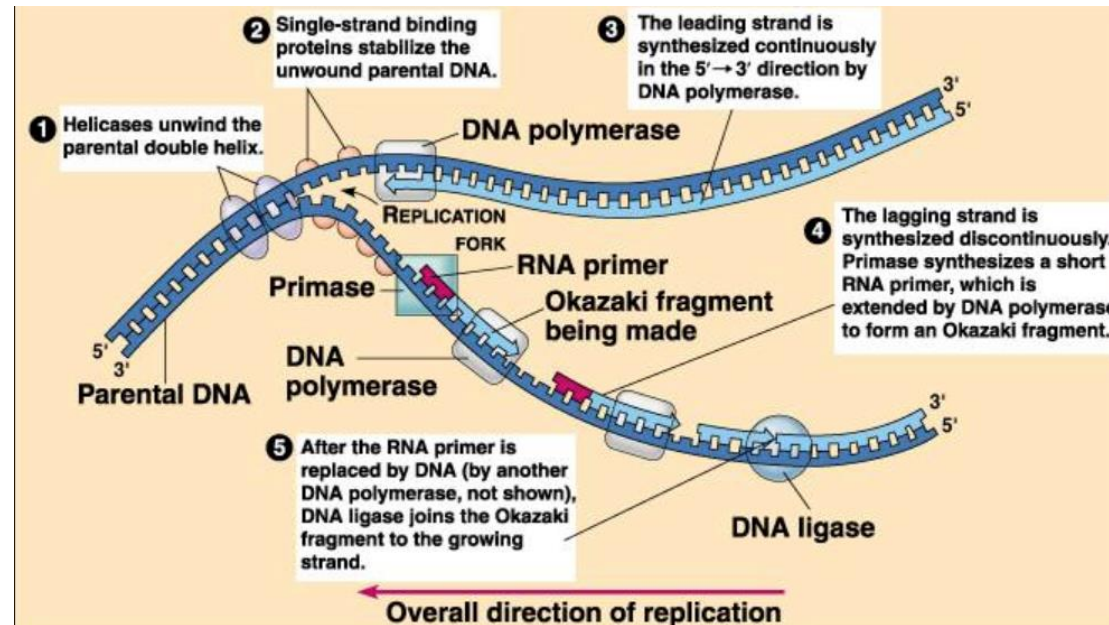
II. Electrophoresis

I. PCR – Polymerase Chain Reaction

Recall...

Replication of genetic materials in living cells

- **Prokaryote, eukaryote**
 - Various enzymes required

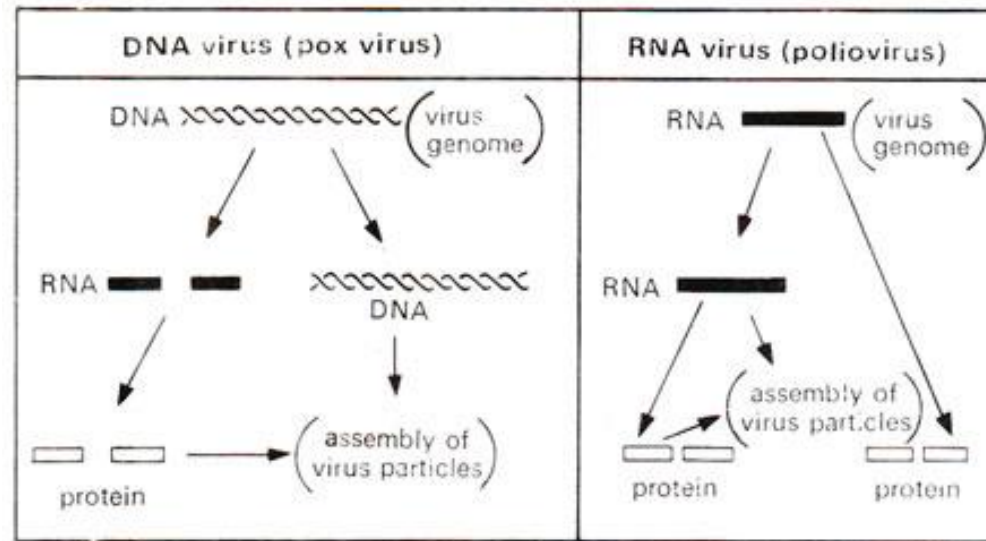


(<https://www.pathwayz.org>)

Recall...

Replication of genetic materials in living cells

- **Virus**
 - DNA virus
 - RNA virus
 - need host cell

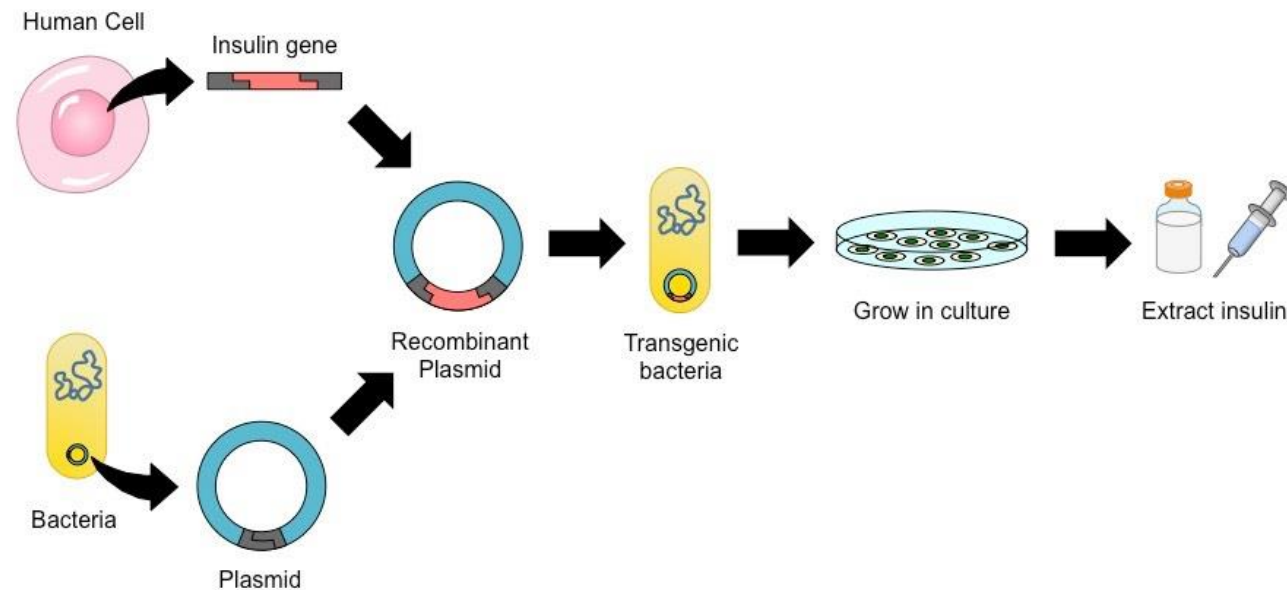


(<https://biocyclopedia.com>)

Replication of genetic materials in vitro/tubes?

– Meaning...

- **Cloning of gene** (details in chapter 3)



Replication of genetic materials in vitro/tubes?

– Meaning...

- **Detections...**
- **Sequencing...**
 - amplification → sequencing

History of PCR

- **Kary Mullis (1944 – 2019)**
 - 1985
 - Nobel Prize in Chemistry in 1993

..the idea of PCR came to him
while driving with his girlfriend
on a highway..

*"It was quiet and something
just went, Click!"*

KARY B MULLIS

1 9 4 4 - 2 0 1 9

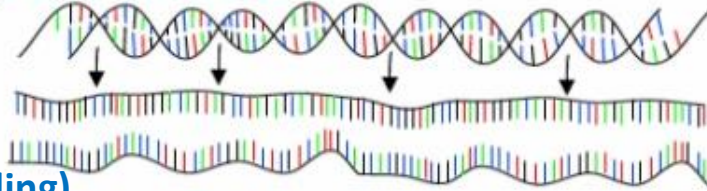
Inventor of PCR Technique



Principles of PCR

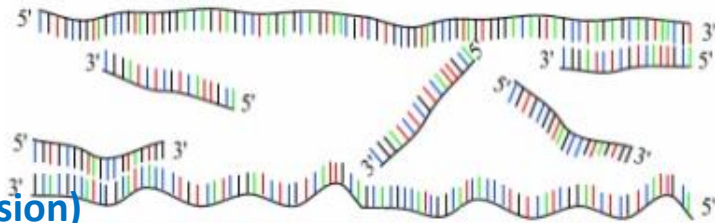
(Denaturation)

Giai đoạn biến tính (92-96°C)



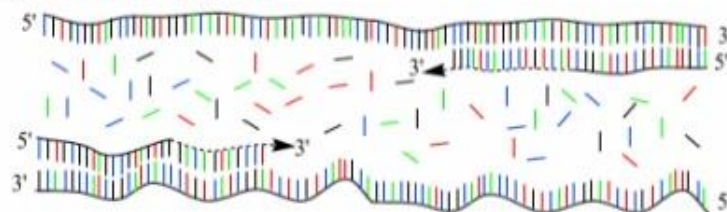
(Annealing)

Giai đoạn bắt cặp (45-55°C)

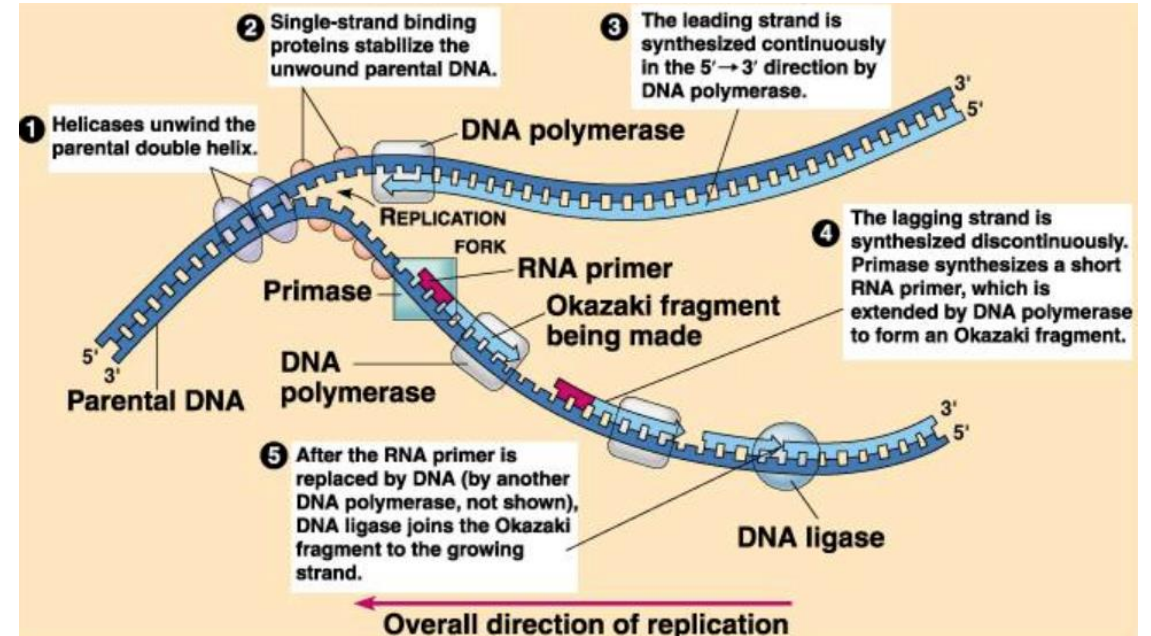


(Extension)

Giai đoạn kéo dài (72°C)



Stages of PCR



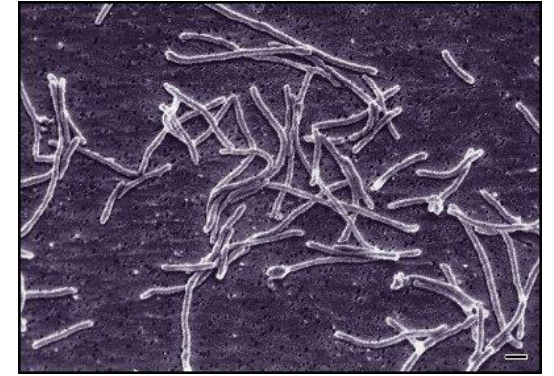
Replication of DNA in living cells

Components in PCR

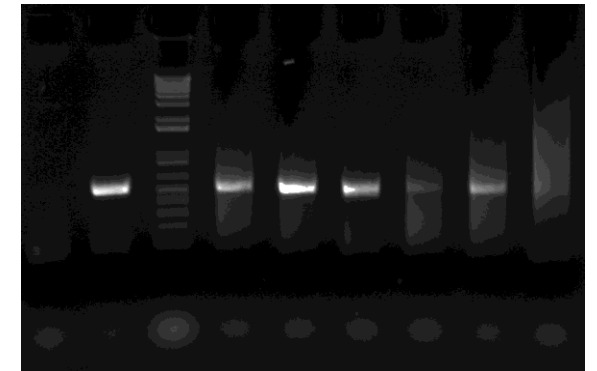
- Enzyme polymerase
- DNA template
- dNTP (dATP, dTTP, dCTP, dGTP)
- Primers (F, R)
- Buffer

DNA polymerase

- First PCR protocol of Kary Mullis: used enzyme DNA polymerase from *E. coli*
- Thermostable enzyme: *Taq polymerase*
- **Advantages:** thermostable, cheap, 'overhang A' in product
- **Disadvantages:**
 - + a high error rate $\sim 10^{-4}$ - 10^{-5}
 - + smear



Thermus aquaticus



Electrophoresis of PCR product with smear

DNA polymerase

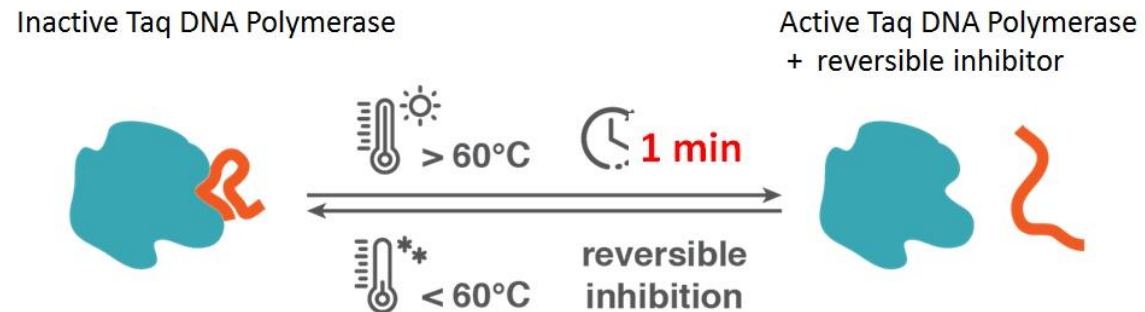
Next-generation DNA polymerases

- high fidelity
~ 10^{-6} - 10^{-7}

- hot-start



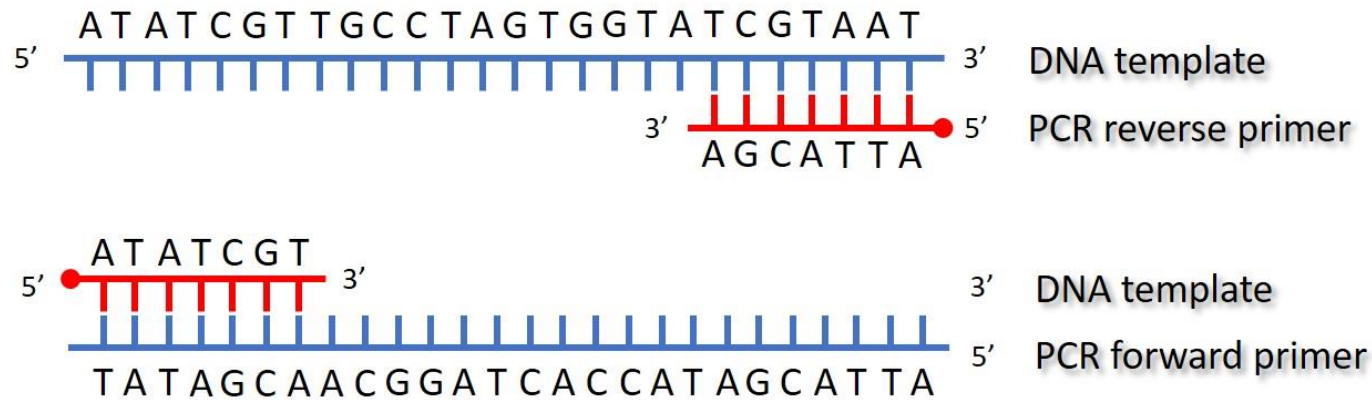
(*Pyrococcus furiosus*)



(<https://www.prima-lifescience.com>)

Primers (F, R)

- Length 20-30 nu
- T_m



- Where to order primers?

Other components

- **DNA template**
 - Nucleic acid extracted (chapter 1)
 - Microbial mass
- **dNTP (dATP, dTTP, dCTP, dGTP)**
 - purchased from suppliers
- **Buffer**
 - purchased from suppliers

Thermal cycler



First version



Modern versions



Clip of PCR

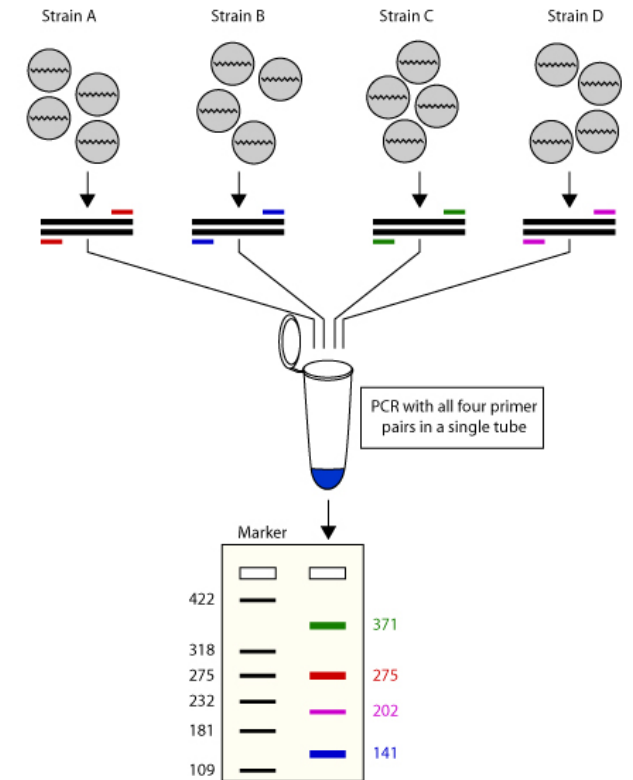
Polymerase chain reaction
(PCR)

PCR methods

1. Monoplex PCR (PCR đơn môi)

2. Multiplex PCR (PCR đa môi)

- Multiplex PCR is a molecular biology technique that allows simultaneous amplification of multiple DNA targets in a single PCR reaction.
- **Advantages**
 - + Saves time, cost, and sample volume
 - + Enables high-throughput detection
- **Limitations**
 - + Requires optimized conditions since multiple primers can compete.
 - + Risk of non-specific amplification or unequal product yield.
 - + More complex primer design compared to conventional PCR.



PCR methods

3. Nested PCR (PCR tổ)

Nested PCR is a modification of the standard PCR technique designed to increase specificity and reduce non-specific amplification.

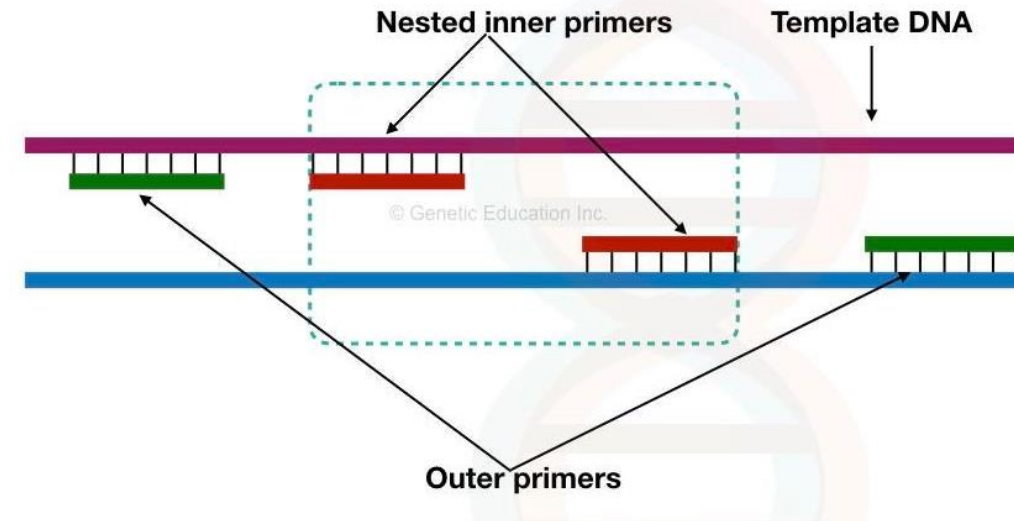
Two rounds

First Round (Outer PCR): Uses an outer primer pair that binds to regions flanking the target DNA.

Second Round (Inner PCR): Uses a second set of primers called inner (nested) primers, which bind inside the first amplicon.

Advantages

- Greatly reduces background noise and false positives.
- Ensures high specificity, useful when the target DNA is present in very low amounts.



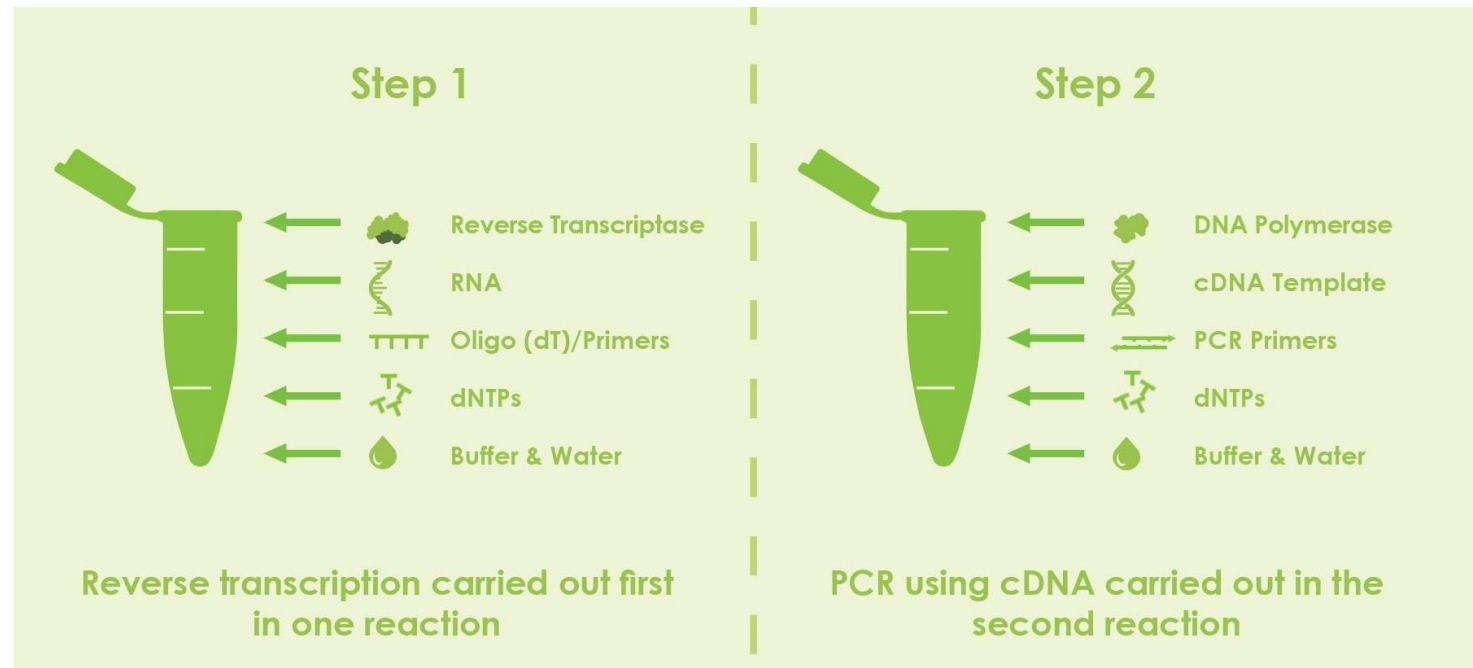
PCR methods

4. RT-PCR

- combines Reverse Transcription with Polymerase Chain Reaction (PCR).
- 2 steps:

Step 1: RNA is converted into complementary DNA (cDNA) using the enzyme reverse transcriptase.

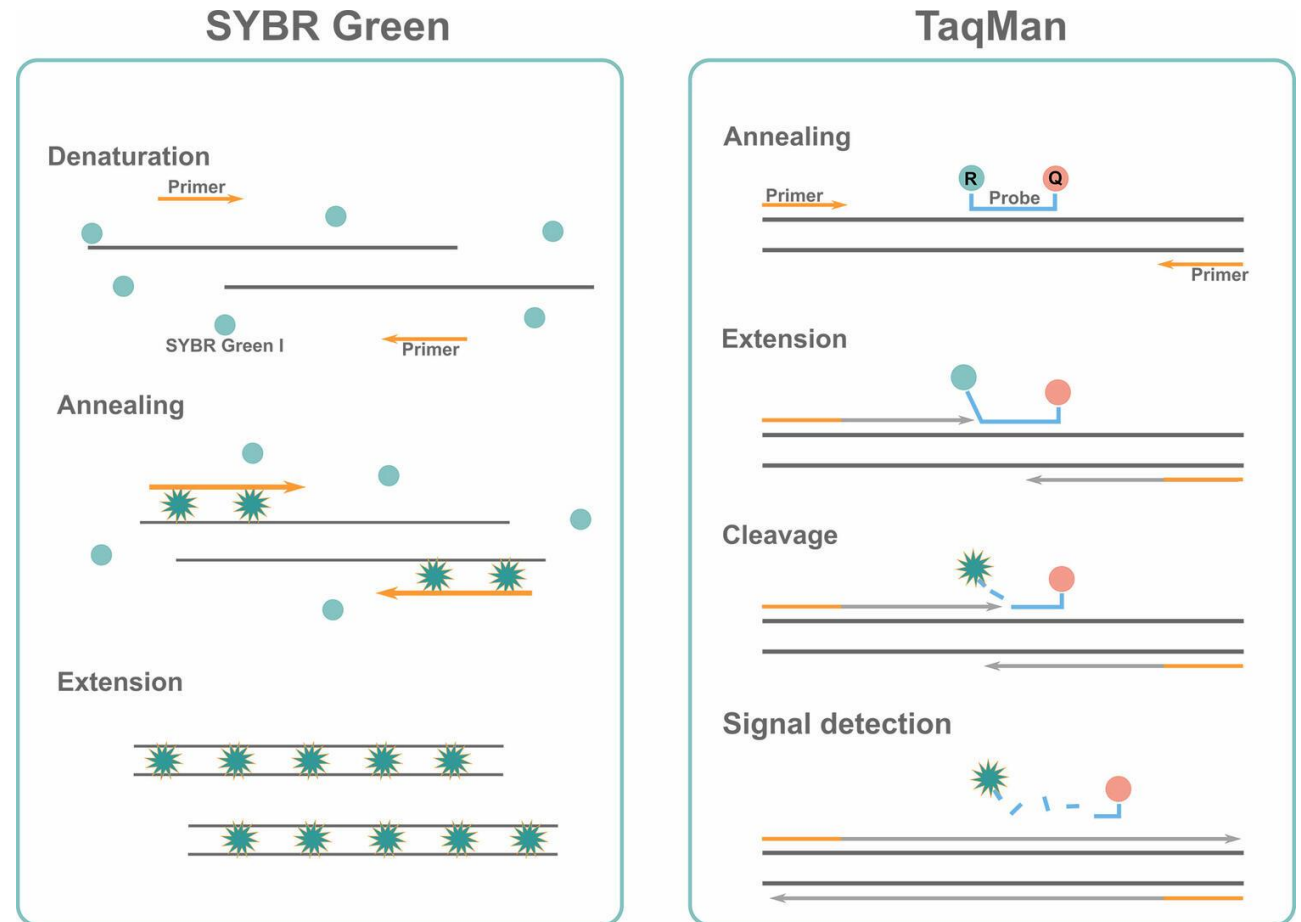
Step 2: The resulting cDNA is then amplified by conventional PCR.



PCR methods

5. Real time PCR (PCR định lượng)

- Real-Time PCR, also known as quantitative PCR (qPCR), allows the detection and quantification of DNA during the amplification process.
- qPCR tracks the reaction in real time using fluorescent signals.
- Fluorescence: Special dyes (e.g., SYBR Green) or probes (e.g., TaqMan) emit fluorescence when bound to DNA.

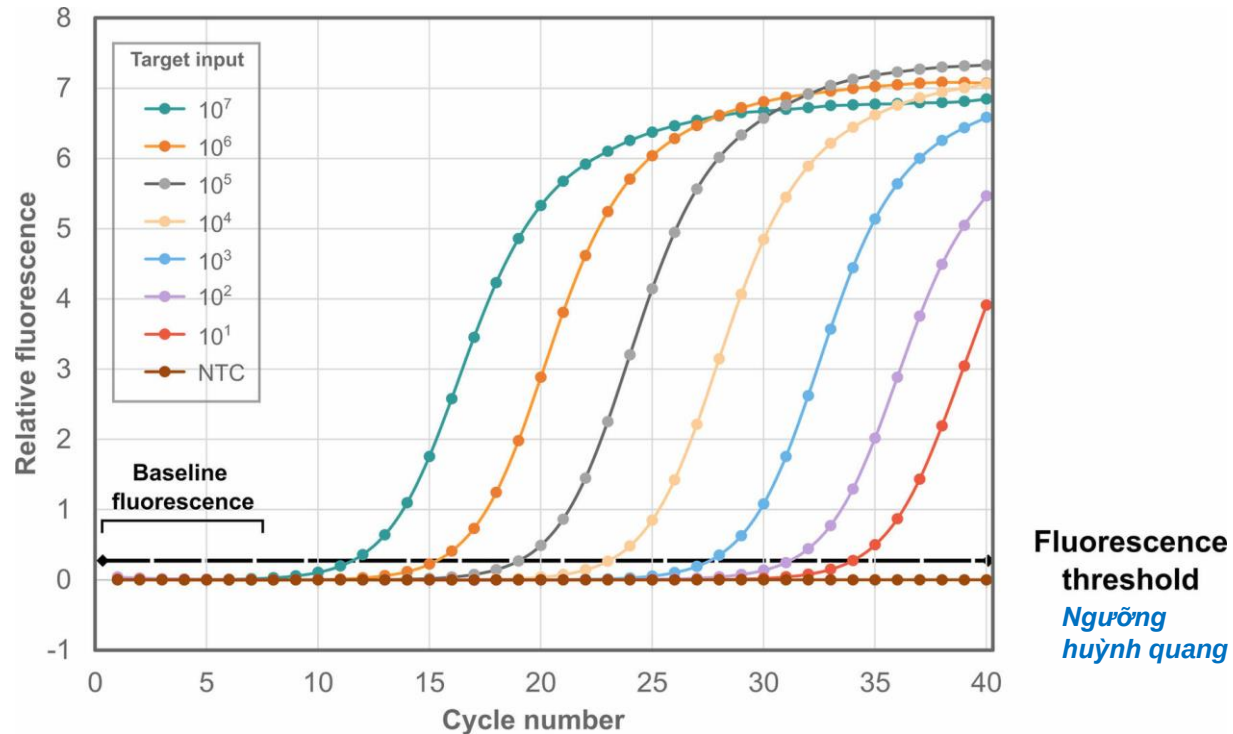


(<https://www.integra-biosciences.com>)

PCR methods

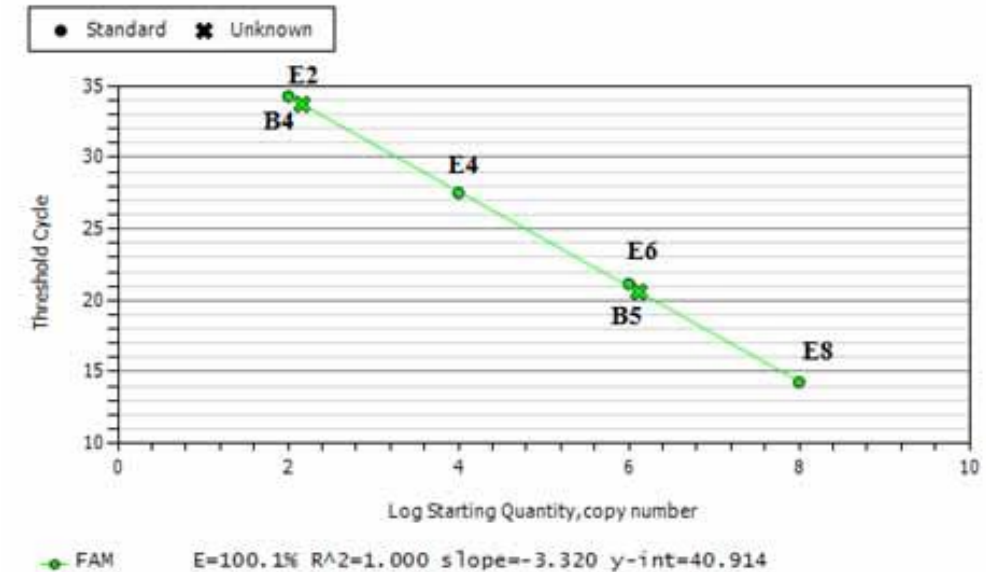
5. Real time PCR (PCR định lượng)

Amplification plot



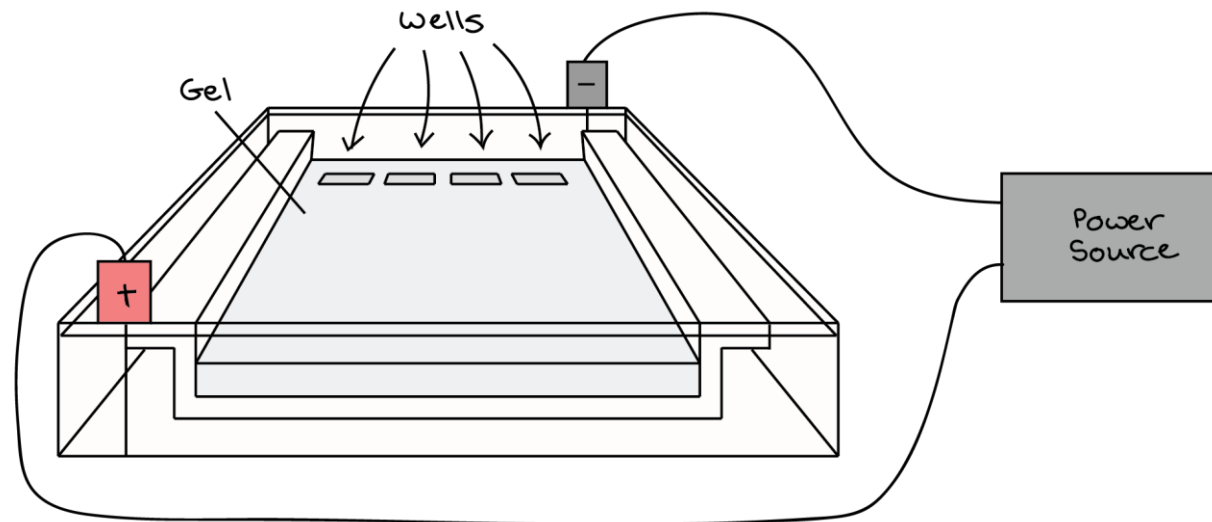
Cycle threshold – Ct (Chu kỳ ngưỡng): cycle number at which the fluorescent signal reaches the fluorescence threshold

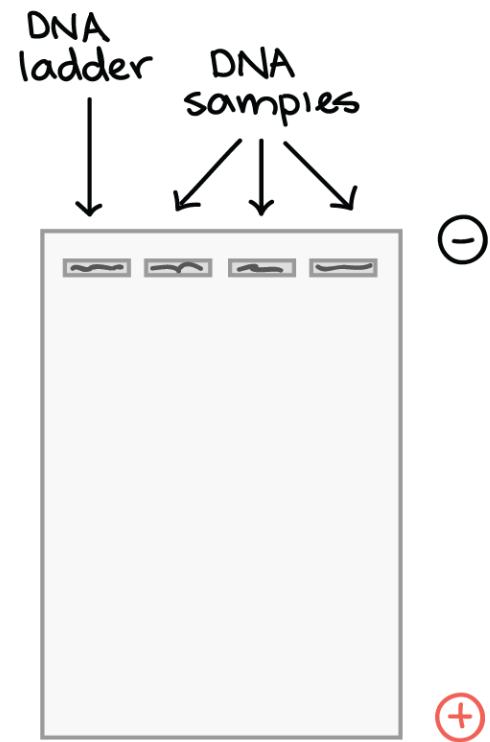
A standard curve of Realtime PCR



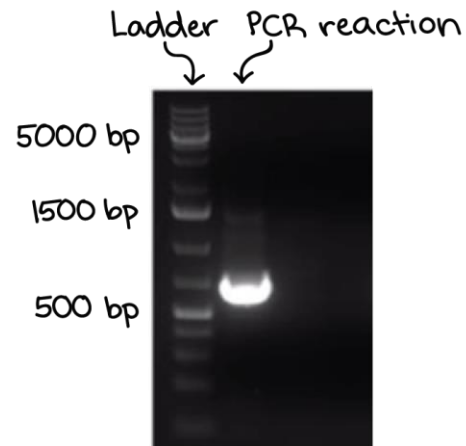
II. ELECTROPHORESIS

- Electrophoresis is a laboratory technique used to separate molecules (DNA, RNA, or proteins) based on their size and charge by applying an electric field.
- Agarose Gel Electrophoresis (AGE): Commonly used for DNA/RNA separation

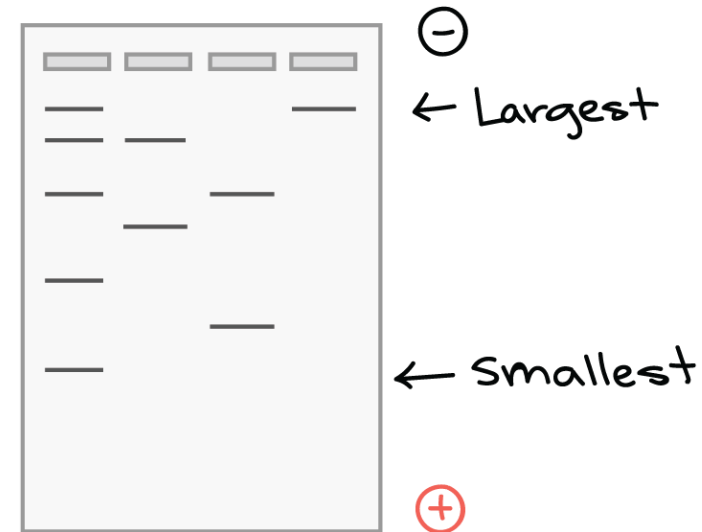




DNA samples are loaded into wells



Power is turned on and DNA fragments migrate through gel



The fragments are now separated by size.

CONCLUSION

- Principles of PCR
- PCR methods
- Principles of gel electrophoresis